

Moein Razavi

AI/ML Engineer | Data Scientist | Software Engineer

McKinney, TX | (979) 676-7486 | razavi.moein94@gmail.com

linkedin.com/in/moein-razavi | github.com/moeinrazavi | Google Scholar

Professional Summary

Data scientist and software developer with a Ph.D. in Industrial Engineering. Develops AI applications, forecasting and energy decision systems at Ford, with additional experience in wearable health software. Builds data pipelines, machine learning models, cloud APIs, and mobile and web interfaces. Published research has 500+ citations on Google Scholar as of September 2026.

Professional Experience

Ford Motor Company

Data Scientist | Aug 2023 - Present

- **Utility data automation:** Built an AI-assisted onboarding pipeline that turns inconsistent customer electricity-usage records into validated data for energy analysis. Replaced utility-specific parsing workflows with reusable schema discovery, extraction, and validation using FastAPI, pandas, Gemini, and BigQuery.
- **EV analytics assistant:** Enabled natural-language questions about vehicle programs, utility rates, and technical documentation through a multi-agent application. Combined LangChain routing, BigQuery text-to-SQL, document retrieval, and an answer-checking step to connect responses with relevant data and domain rules.
- **Vehicle-to-grid forecasting and dispatch:** Built software to estimate available fleet energy and select vehicles for grid commitments. Combined transformer forecasts, market prices, and Monte Carlo simulation to account for uncertainty, with load-reallocation logic for vehicles that disconnect.
- **Forecasting models:** Trained Temporal Fusion Transformer and Autoformer models with Optuna tuning on Vertex AI GPUs, comparing vehicle-availability forecasts with a previous-week baseline to support energy planning.
- **Rate recommendation:** Developed an application that uses XGBoost electricity-bill predictions, vehicle telemetry, and household consumption to inform utility-plan recommendations. Built Flask APIs, BigQuery workflows, and a Dash interface with SAML authentication; deployed on Cloud Run through Tekton CI/CD.
- **Energy analytics infrastructure:** Automated preparation of energy-simulator inputs and outputs using PySpark, BigQuery, and Airflow. Built discharge-analysis dashboards and REST APIs to make vehicle energy activity and operational performance available to business teams.
- **Code analysis and cloud cost estimation:** Built a FastAPI tool that analyzes repository code, suggests optimizations, and estimates compute costs using reference pricing. Combined Python AST parsing with LLM analysis and streamed findings to the browser as processing completed.

CereVu Medical, Inc.

Software Developer / Research Engineer (Contract) | Jun 2023 - Sep 2024

- Developed Vitality and Clinical, 2 SwiftUI applications connecting wearable sensors with patient and clinician workflows. Integrated CoreBluetooth and Swift Charts to acquire and display 9 physiological signals.
- Migrated the Clinical app from UIKit to modular SwiftUI components using MVVM-Cordinator; implemented role-based views, sensor calibration, data export, and authentication.
- Built Python/Flask services for biosignal streaming and a React/Next.js dashboard backed by MongoDB. Deployed containerized backend services on AWS EC2 with GitHub Actions and delivered iOS builds through TestFlight.

Education

Ph.D., Industrial Engineering | Texas A&M University | Dec 2023

Master of Engineering, Computer Engineering | Texas A&M University | May 2021

Selected Projects

- **Business Action Prioritizer:** Built a Google ADK application that ranks business actions against a scoring rubric, using PandasAI for analysis, human approval of the plan, and an MCP tool to publish reports to Cloud Storage.
- **Inspecalytics:** Built a field-inspection platform with offline data collection, photo annotation, and automated Word/PowerPoint reports. Connected SwiftUI and web interfaces through Supabase synchronization, with Fastlane/GitHub Actions release automation.
- **Clink Social:** Developed a SwiftUI application with messaging, location-based discovery, and a personalized feed, backed by Firebase and TypeScript cloud functions. Implemented QR-based cross-device sign-in using time-limited token exchange.

Research Experience

Texas A&M University

Research Assistant | Sep 2018 - May 2024

- **Wearable stress detection:** Developed and evaluated models using heart-rate and motion data from 54 students over 40 days, with self-reported stress labels and analysis of predictive features.
- **MERCURY training acceleration:** Co-authored computation-reuse research reporting 1.97x average training speedup across 12 neural networks with accuracy similar to its baseline.
- **OpenSync:** Co-developed an open-source Python platform synchronizing neuroscience sensors and experiment events through Lab Streaming Layer; published benchmarks measured below 0.07 ms RMS recording-time lag.
- **Computer vision and health applications:** Developed video-based distance and face-mask monitoring, plus GlucoseCoach, an iOS/Flask application for personalized glucose monitoring.

Technical Skills

Languages: Python, SQL, Swift, TypeScript, JavaScript, C++, Java, R, MATLAB

Machine Learning: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face, Optuna, OpenCV, time-series forecasting, classification, model validation, SHAP/LIME

Generative AI: LangChain, LangGraph, Google ADK, OpenAI Agents SDK, Gemini, RAG, FAISS, ChromaDB, prompt engineering, text-to-SQL, MCP

Data: pandas, NumPy, SciPy, PySpark, Airflow, ETL/ELT, PostgreSQL, MongoDB, Redis, Firebase, Supabase

Applications: FastAPI, Flask, REST APIs, WebSocket, React, Next.js, SwiftUI, UIKit, CoreBluetooth, MVVM, OAuth 2.0, JWT, SAML

Cloud: AWS (S3, EC2, Lambda, SageMaker), GCP (Vertex AI, BigQuery, Cloud Run, Dataproc), Azure OpenAI

Delivery: Git, Docker, Kubernetes, Terraform, CI/CD, Tekton, GitHub Actions, AWS CodeDeploy, Fastlane, TestFlight, XCTest

Selected Publications

Razavi, M., et al. Machine Learning, Deep Learning, and Data Preprocessing Techniques for Detecting, Predicting, and Monitoring Stress and Stress-Related Mental Disorders: Scoping Review. JMIR Mental Health, 2024.

Janfaza, V., Weston, K., Razavi, M., et al. MERCURY: Accelerating DNN Training By Exploiting Input Similarity. IEEE HPCA, 2023.

Razavi, M., et al. OpenSync: An open-source platform for synchronizing multiple measures in neuroscience experiments. Journal of Neuroscience Methods, 2022.

Razavi, M., et al. An Automatic System to Monitor the Physical Distance and Face Mask Wearing of Construction Workers in COVID-19 Pandemic. SN Computer Science, 2022.